

Unsupervised Discovery of Market Regimes Using Static and Temporal Models

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Abstract

How persistent are unsupervised market regimes to an analyst who cannot see the future? I treat this as a measurement question and audit the look-ahead content of regime persistence in a head-to-head comparison of a static Gaussian mixture model (GMM) and a temporal hidden Markov model (HMM), fit to identical weekly macro-financial data—the S&P 500 return, the VIX, and the 10Y–2Y yield spread, 1994–2023—with the same number of states and evaluation criteria, in a fully reproducible pipeline in which every published number is verified against archived outputs. Look-ahead enters through three separable channels—standardization, decoding, and parameter estimation—that affect the two models asymmetrically: only the HMM has a decoding step, so the pointwise GMM serves as a control for the channel where the temporal model is most exposed. Causal decoding alone cuts the HMM’s apparent mean regime duration by about a quarter, and the HMM-to-GMM duration ratio falls from $6.2\times$ under the naive configuration to $2.2\times$ under a rolling causal standardization—but only to $5.2\times$ under an expanding one, so the size of the correction depends on which causal scheme is adopted. A twenty-seed restart diagnostic shows the fitted EM optimum moves these figures by more than either corrected channel, making initialization a further choice that must be disclosed. The models substantially disagree on the partition (adjusted Rand index 0.33–0.35), with agreement concentrated in market stress. The regimes’ economic content—alignment with NBER recessions and a $VIX > 20$ threshold—survives the full-sample and rolling corrections and weakens under the expanding one: the look-ahead bias lies mainly in the regimes’ claimed persistence, not in their existence.

1 Introduction

Financial markets move through persistent phases—calm expansions, volatile corrections, acute crises—that differ in their risk and return characteristics. Identifying these market regimes is a long-standing problem in empirical finance, with direct consequences for allocation and risk management [1, 6], and unsupervised learning offers an appealing approach to it: rather than imposing regimes through predefined rules such as recession dates or volatility thresholds, one lets them emerge from the data. The central modelling decision is then how much temporal structure to build in. A static clustering method treats each period independently and asks only which latent state its features resemble; a temporal model additionally parameterizes how states persist and transition over time. Gaussian mixture models (GMM) and hidden Markov models (HMM) are the standard representatives of these two choices.

The usual argument for the temporal model is persistence: HMM regimes are reported to last on the order of economic cycles, where static methods flicker from week to week [7]. Yet a reported regime duration is a property of a decoded label sequence computed from standardized inputs, and

both steps admit choices that quietly condition on the future. The most-likely-path (Viterbi) and smoothed decoders label each week using the entire observation sequence, including its future; only the forward-filtered decoder restricts itself to the past. The same kind of look-ahead enters when features are z -scored with full-sample statistics. These choices are routinely left unstated, and they are not innocuous—a regime duration measured with future information is not the duration an analyst could have observed in real time. The primary object of this paper is therefore not the models but the measurement: how much of the HMM’s celebrated persistence is look-ahead?

To answer this, the comparison is run under controlled conditions. Both models are fit to the same weekly panel of three macro-financial features over 1994–2023, with the same number of states and the same evaluation criteria, in a pipeline that verifies every published number against archived outputs. The organizing idea is that look-ahead enters through three separable channels—standardization (L1), decoding (L2), and parameter estimation (L3)—which have different corrections and, crucially, affect the two models asymmetrically: the HMM’s decoding step has no analogue in the static GMM, which labels each week from that week’s features alone. The GMM is therefore a built-in control for exactly the channel where the temporal model is most exposed, and separating the channels makes it possible to attribute the persistence gap to its sources rather than reporting it as a single headline number.

The persistence advantage of the HMM is real in direction—it survives every configuration tested—but its magnitude is a function of the measurement. Under the naive configuration (full-sample standardization, Viterbi decoding) the HMM’s regimes last $6.2\times$ longer than the GMM’s. Causal decoding alone cuts the HMM’s apparent duration by about a quarter, from 55.9 to 41.2 weeks. The fully causal endpoint, however, depends on which causal standardization is adopted: a rolling scheme yields a ratio of $2.2\times$, an expanding scheme $5.2\times$, and the bootstrap intervals of the two do not overlap. The divergence is informative rather than embarrassing—Section 2.2 argues, and Section 4.2 confirms, that rolling re-normalization does not merely remove future moments but redefines each feature relative to its trailing year, a related but different economic object.

A second finding concerns estimation rather than information. The Baum–Welch likelihood is multimodal, and a twenty-seed diagnostic shows that single-run fits scatter across local optima whose decoded partitions differ substantially—adjusted Rand indices as low as 0.32 between restarts. An earlier version of this study rested on a fit 86 nats below the best optimum, and that difference alone moved the headline figures by more than either look-ahead correction. Initialization is not a look-ahead channel—it is estimation variance, not information leakage—but it belongs to the same class of unstated pipeline choices, so the models here are fit with multiple restarts and the protocol disclosed (Section 4.6).

None of the individual ingredients is new. The distinction between filtered and smoothed inference is textbook [9], Markov-switching models of macro-financial states are standard [4], look-ahead bias is a familiar hazard, and the sensitivity of EM to initialization is well documented [10]. What this paper contributes is the disciplined, reproducible execution of a comparison usually made informally: identical conditions, the look-ahead channels separated, the estimation protocol surfaced, and the persistence claim corrected rather than asserted. The remainder of the paper describes the data and features (Section 2), the models and the look-ahead taxonomy (Section 3), and the results (Section 4), before discussing what survives the corrections (Section 5) and what the analysis does not establish (Section 6).

Table 1: Feature set. All series are sampled at the Friday close (W-FRI).

Feature	Definition	Source
SP500_Return	Weekly log return of the S&P 500 closing level	Yahoo Finance
VIX	CBOE Volatility Index, weekly level	Yahoo Finance
Yield_Spread	10Y minus 2Y Treasury constant-maturity yield	FRED

2 Data and Features

The study uses a single weekly panel of three macro-financial series. Observations are sampled on Fridays (pandas frequency W-FRI) and span 14 January 1994 – 29 December 2023, giving 1,564 weekly observations across roughly three decades of equity-market history, covering the dot-com bubble, the 2007–2009 financial crisis, and the COVID-19 crash. All series are drawn from public sources,¹ so the dataset can be reconstructed from public sources by anyone without licensed data (Appendix A notes one caveat).

2.1 Features

Each weekly observation is described by three features spanning the equity, volatility, and macroeconomic dimensions of market conditions (Table 1). The S&P 500 weekly log return captures equity-market direction; the CBOE Volatility Index (VIX) measures forward-looking implied volatility and risk appetite; and the 10-year – 2-year Treasury yield spread summarizes the slope of the yield curve, a standard business-cycle indicator. The feature set is deliberately compact: three interpretable inputs are chosen to span distinct economic channels rather than engineered for predictive power, consistent with the descriptive, unsupervised aim of this work.

2.2 Standardization

All models are fit on standardized features, so that no single series dominates the Gaussian covariance geometry through its raw scale—VIX levels and yield spreads occupy very different numerical ranges. The choice of how to standardize is not harmless: it determines whether the inputs at week t depend only on information available by week t , or on the full sample. Three schemes are considered (Table 2).

The full-sample scheme computes each feature’s mean and standard deviation over the entire sample and applies them uniformly.² This is the reference baseline—it reproduces the archived regime labels exactly (Section 4.1)—and it is the textbook default, but it is non-causal by construction: the z -score at any week embeds the sample mean and variance, which depend on all weeks, including future ones. The two causal alternatives remove this dependence. Rolling-52 standardizes each week using only the trailing 52-week window; expanding uses all weeks up to and including the current one, subject to a 52-week minimum. Both require a 52-week history and therefore discard a 51-week burn-in, retaining 1,513 weeks.

¹S&P 500 and VIX from Yahoo Finance (<https://finance.yahoo.com>; tickers ^GSPC and ^VIX); the 10-year minus 2-year Treasury constant-maturity spread from the Federal Reserve Economic Data database (FRED, <https://fred.stlouisfed.org>).

²All three schemes use the population standard deviation (ddof = 0), matching scikit-learn’s `StandardScaler`. An earlier version of this pipeline used the sample estimator (ddof = 1) for the two causal schemes; the estimators are unified in this revision.

Table 2: Standardization schemes. Causal variants drop a 51-week burn-in.

Scheme	Window	Causal	Weeks retained
Full-sample	Entire sample	No	1,564
Rolling-52	Trailing 52 weeks	Yes	1,513
Expanding	All weeks to date (≥ 52)	Yes	1,513

The two causal schemes are not interchangeable, and the distinction matters for everything that follows. Expanding standardization is the causal counterpart of the full-sample scheme: its estimates converge to the full-sample moments, so late-sample inputs are nearly identical under the two, and correcting with it isolates the removal of future information. Rolling standardization does something more: it re-normalizes each feature relative to its own trailing year, so that a VIX of 22 after a placid year is coded as extreme while the same level after a turbulent year is unremarkable. That is a different economic object—local abnormality rather than absolute level—not merely a bias-corrected version of the same one. Sections 4.2 and 4.4 show the consequences: the two causal schemes lead to materially different persistence and validation figures, for reasons this distinction predicts.

This standardization channel is the first of three distinct routes through which look-ahead information can enter the pipeline; the full taxonomy is developed in Section 3.3. For economic interpretation, regime profiles in Section 4.5 are reported in raw (unstandardized) units, while all model fitting uses standardized inputs.

3 Methods

Two unsupervised models partition the weekly feature panel into three regimes, deliberately spanning a spectrum of temporal assumptions: a static Gaussian mixture that treats each week independently, and a hidden Markov model that explicitly parameterizes regime persistence. Both use three components and full covariance matrices, and both are evaluated on identical data, features, and criteria. Fixing the number of regimes at three (Calm, Transitional, Stressful) keeps the models directly comparable; sensitivity to this choice is discussed in Section 6.

3.1 Models

Gaussian mixture model (GMM). The GMM treats each weekly feature vector as an independent draw from a mixture of three multivariate Gaussians and is fit by expectation–maximization using `scikit-learn` [8].³ Each week is then assigned to the component with the highest posterior responsibility. This assignment is pointwise: the label at week t depends only on that week’s features and the fitted parameters, never on neighbouring weeks. The GMM therefore has no notion of regime persistence and no decoding step—a fact that becomes central below.

Hidden Markov model (HMM). The HMM augments the same three-component Gaussian emission structure with a first-order Markov chain over the latent regime, estimating a 3×3 transition matrix A alongside the emission parameters by Baum–Welch. Modelling transitions explicitly is what lets the HMM express persistence: the off-diagonal entries of A penalize switching, so a latent sequence is preferred when it does not flicker. Because the Baum–Welch objective is

³`sklearn.mixture.GaussianMixture: n_components=3, covariance_type="full", init_params="k-means++", n_init=10, random_state=42.`

multimodal and a single EM run can terminate at a poor local optimum [10], the HMM is fit from ten random initializations and the highest-likelihood fit is retained;⁴ the sensitivity being controlled here turns out to be large (Section 4.6). Recovering the regime sequence from the fitted model requires a decoding step (Section 3.2), and the choice of decoder drives a substantial share of the persistence gap reported in informal comparisons. The use of a Markov-switching latent state for macro-financial regimes follows [4].

Regime naming. Both fitted models return unordered latent states; raw indices are not comparable across models, standardizations, or random seeds (label switching). To obtain stable, interpretable names, the states of each fit are sorted by ascending mean VIX: the lowest-VIX state is labelled Calm, the highest Stressful, the remainder Transitional. Because the ordering uses the emission-component means—model parameters, not decoded paths—naming is identical across the three HMM decoders.

Experimental grid. Each model is run under the three standardizations of Section 2.2. The GMM yields one label sequence per standardization; the HMM yields three, one per decoder. The full-sample standardization with the archived labels is the reference configuration.

3.2 Decoding the HMM

Given a fitted HMM, three standard decoders recover a regime for each week, and they differ in exactly one respect: how much of the observation sequence each is allowed to use [9].

- **Filtered (causal):** $\hat{s}_t^{\text{filt}} = \arg \max_i P(s_t = i \mid x_{1:t})$, using only observations up to and including week t .⁵
- **Smoothed (non-causal):** $\hat{s}_t^{\text{smo}} = \arg \max_i P(s_t = i \mid x_{1:T})$, using the entire sequence to label each week.
- **Viterbi (non-causal):** $\hat{s}_{1:T}^{\text{vit}} = \arg \max_{s_{1:T}} P(s_{1:T} \mid x_{1:T})$, the single most-likely path given all T observations.

The filtered decoder is the only online estimate of what an analyst could have known in real time. Both smoothed and Viterbi condition each week’s label on future observations. This is the main point: any persistence a non-causal decoder reads into the sequence by leaning on future data is unavailable to a real-time user and inflates apparent regime duration relative to what the model delivers online.

3.3 Look-ahead taxonomy

Three independent channels can let information from the future enter the regime labels. Separating them is the methodological core of this paper, because they have different fixes and—critically—affect the two models asymmetrically (Table 3).

L1 (standardization). The full-sample z -score uses a mean and variance computed over every week, so each standardized input embeds future information (Figure 1). The causal standardizations

⁴`hmmlearn.hmm.GaussianHMM: n_components=3, covariance_type="full", n_iter=1000, random_state ∈ {42, ..., 51}`, selecting the fit with the maximal total log-likelihood. Section 4.6 quantifies the sensitivity this protocol addresses.

⁵The filtered sequence is the pointwise argmax of the filtered marginals and need not form a transition-consistent path. For the run-length statistics used here this is immaterial, and it is the natural online counterpart of the smoothed argmax.

Table 3: The three look-ahead channels, their causal corrections, and which models they affect.

Channel	Mechanism	Causal fix	Affects
L1 — Standardization	z -score uses full-sample mean and variance	rolling-52 / expanding	GMM, HMM
L2 — Decoding	Viterbi/smoothed condition on $x_{1:T}$	forward-filtered decoding	HMM only
L3 — Estimation	parameters fit once on the full sample	walk-forward refit (not implemented)	GMM, HMM

of Section 2.2 remove it. This channel affects both models equally—though, as that section notes, the rolling variant also changes what the features measure.

L2 (decoding). Non-causal decoders condition each label on the full sequence; forward filtering removes this. The channel is HMM-only: the GMM labels each week pointwise and has no decoding step, so its durations are already causal in this respect. The HMM’s headline persistence under Viterbi is therefore partly an L2 artifact, and the gap between Viterbi and filtered run-lengths measures it directly.

L3 (parameter estimation). Both models are fit once on the full sample, so the estimated parameters embed the entire period; a fully causal protocol would re-fit on a walk-forward basis. This is not implemented and is disclosed in Section 6. Because it is common to both models, it does not bias the GMM–HMM comparison in either direction; it raises the absolute confidence of both, not the contrast between them.

One further source of variation is deliberately excluded from this taxonomy. The restart sensitivity documented in Section 4.6 is estimation *variance*—the fitted optimum depends on the initialization—rather than leakage of future information, and its remedy (multiple restarts with a disclosed selection rule) differs accordingly. It is treated as a companion finding, not a fourth channel.

3.4 Evaluation metrics

Evaluation emphasizes regime persistence, cross-model agreement, and economic interpretability; no predictive metrics are used, consistent with the descriptive aim.

Regime duration. Persistence is measured two ways. The empirical run-length is the mean number of consecutive weeks the label remains unchanged, read directly from a label sequence; for the HMM it is decode-sensitive. The analytic expected duration of state i follows from its self-transition probability under the fitted model, $\mathbb{E}[\text{dur}_i] = 1/(1 - A_{ii})$ (geometric dwell time); it depends only on A and is therefore decode-invariant, and it is what the regime-switching literature usually means by “expected duration.” The geometric dwell distribution is itself an assumption of the first-order chain—one that hidden semi-Markov models exist to relax [2]—so the analytic figure should be read as model-implied, not model-free. Reporting both isolates how much decoded persistence comes from the model’s transition structure versus from the decoder’s use of future data. The GMM has no model transition matrix, so its persistence is summarized by the empirical run-length and an empirical transition matrix estimated from its label sequence.

Uncertainty. A mean run-length is a mean over spells, so uncertainty is quantified with a spell bootstrap: the observed spells of a label sequence are resampled with replacement (10,000 draws) and 95% intervals are read from the percentiles of the resampled means. Intervals for duration *ratios* divide the bootstrap draws of the two models elementwise, treating the two spell samples

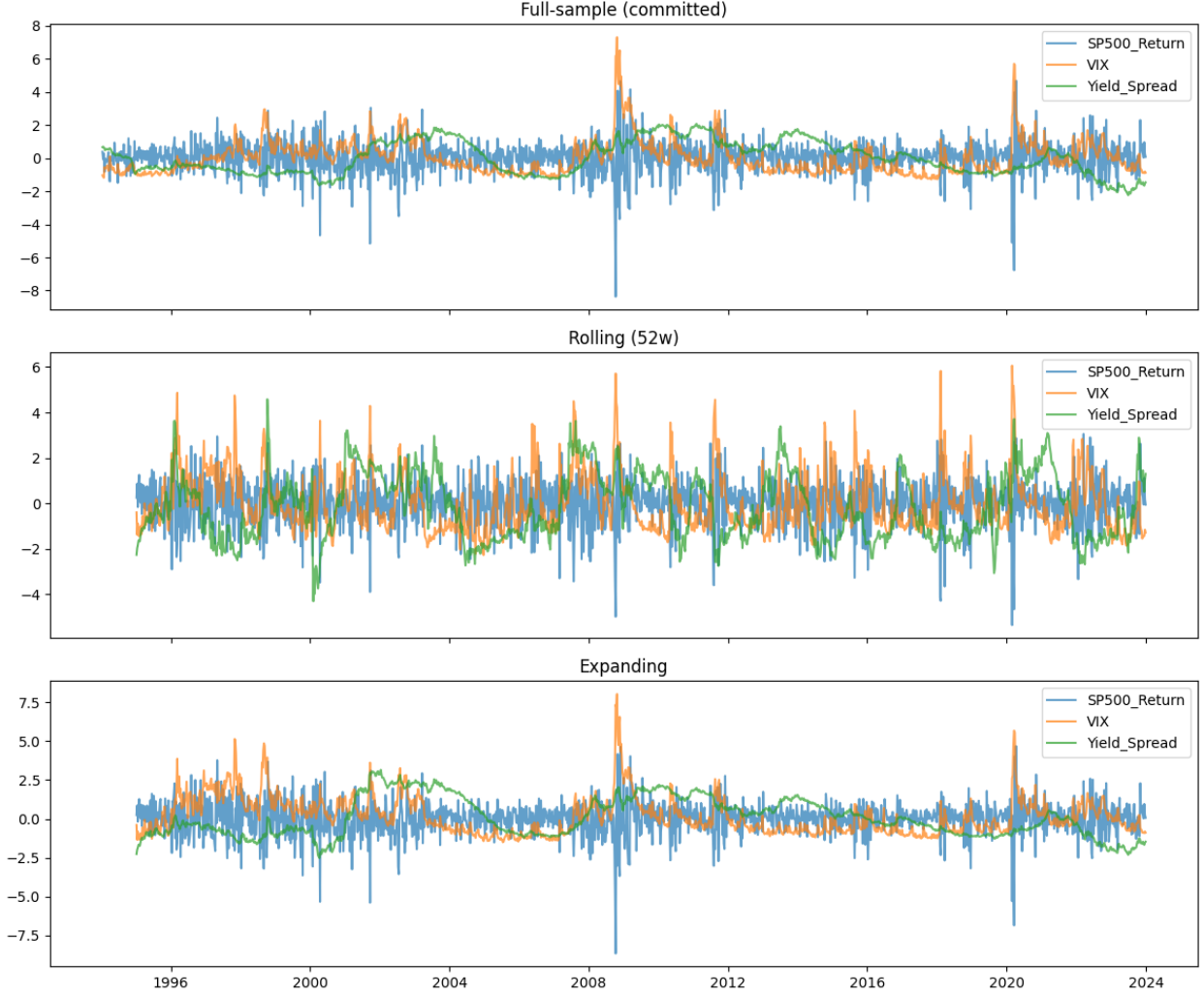


Figure 1: Standardized features under the full-sample, rolling-52, and expanding schemes (channel L1).

as independent—an approximation, since both models are fit to the same weeks, and one that is disclosed rather than corrected here.

Decode look-ahead share. To quantify L2 directly, I report the fraction of weeks whose HMM label changes between the non-causal (Viterbi) and causal (filtered) decoders: the share of the labelling that depends on future information.

Transition entropy. For each regime, the predictability of the next regime is the Shannon entropy in bits of the corresponding transition-matrix row, $H_i = -\sum_j P_{ij} \log_2 P_{ij}$, with P estimated empirically from the label sequence so that the GMM and each HMM decode are treated identically; lower entropy means a more predictable regime process. States are mapped to their VIX-ordered names before estimation, so entropies are reported per named regime and are comparable across models.

Cross-model agreement. Agreement between the GMM and HMM label sequences is measured by the adjusted Rand index [5], which corrects raw cluster overlap for chance and is invariant to label permutation. It is reported for the full sample, split pre/post-2005, and checked for sensi-

Table 4: Mean regime duration in weeks, by standardization and HMM decoder. Brackets are 95% spell-bootstrap intervals (Section 3.4); the full-sample HMM estimates rest on only 28–38 spells, hence their width.

Standardization	GMM	HMM Viterbi	HMM smoothed	HMM filtered
Full-sample	9.0 [6.6, 11.6]	55.9 [35.2, 80.4]	55.9 [35.2, 80.4]	41.2 [24.7, 60.2]
Rolling-52	6.2 [5.2, 7.2]	18.9 [14.9, 23.4]	17.2 [14.1, 20.6]	13.5 [11.2, 16.1]
Expanding	7.3 [5.5, 9.3]	50.4 [31.3, 73.8]	50.4 [31.3, 73.7]	37.8 [22.9, 56.8]

tivity to the HMM decoder.

3.5 External validators

Because the regimes are unsupervised, their economic content is tested against two external references that play no part in fitting.

NBER recessions. Each week is classified as recession or expansion using the official NBER business-cycle chronology.⁶ The sample contains three recessions: the 2001 dot-com downturn, the 2007–2009 financial crisis, and the 2020 COVID-19 contraction—134 recession weeks in total. A regime set with genuine economic content should concentrate its Stressful label in recession weeks; I report the share of recession and non-recession weeks labelled Stressful, overall and per episode.

VIX threshold concordance. As a second, model-free reference, weeks are flagged by the conventional elevated-volatility rule $VIX > 20$ (607 weeks, 38.8% of the sample). Concordance between the Stressful label and this indicator is summarized by Cohen’s κ [3], with the recall and precision of the Stressful label against the threshold. This tests whether the unsupervised Stressful regime recovers a standard volatility cut-off without being told to.

4 Results

4.1 Reproduction

Before any analysis, the pipeline verifies itself at two levels. A *label gate* refits both models on the archived full-sample standardized features and requires the decoded labels to match the archived label files; both models reproduce exactly (adjusted Rand index 1.000). A *numbers gate* extends the same discipline to every table in this paper: each is written to the repository once, and every subsequent run must reproduce it to within 10^{-6} , so a stale or externally computed figure fails loudly instead of drifting silently out of sync with the pipeline. The published numbers have additionally been reproduced in an independent environment under the pinned software stack of Appendix A. All numbers below are computed from this verified baseline.

4.2 Persistence under causal measurement

The headline contrast of informal comparisons—HMM regimes lasting roughly a year against the GMM’s few weeks—is partly real and partly an artifact of how the HMM sequence is standardized and decoded. Table 4 decomposes mean regime duration across the three standardizations and, for the HMM, the three decoders, with spell-bootstrap intervals throughout.

⁶National Bureau of Economic Research, US business cycle expansions and contractions, <https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions>.

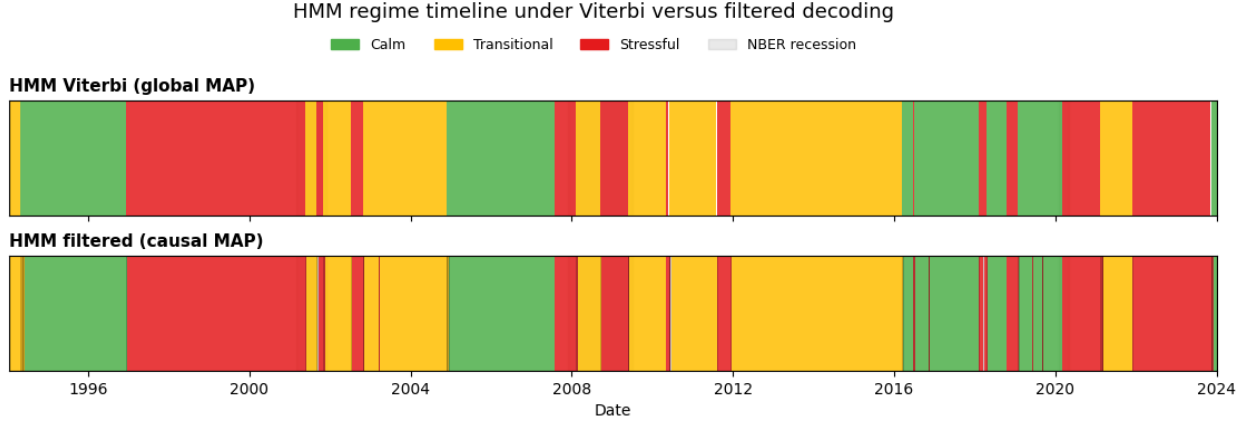


Figure 2: HMM regime timeline under Viterbi versus filtered decoding.

Two readings of the table matter. Under the naive configuration—full-sample standardization with Viterbi decoding—the HMM averages 55.9 weeks against the GMM’s 9.0, a $6.2\times [3.7, 9.9]$ persistence ratio that recovers the “year-long regimes” claim. Under a fully causal configuration the answer splits: rolling standardization with filtered decoding gives 13.5 against 6.2 weeks, a ratio of $2.2\times [1.7, 2.8]$, while expanding standardization with filtered decoding gives 37.8 against 7.3, a ratio of $5.2\times [3.0, 8.5]$. The two intervals do not overlap: how much of the persistence gap survives causal measurement depends materially on which causal standardization is adopted, for reasons Section 2.2 anticipates and Table 6 makes precise. The qualitative ordering survives every cell of the table—the HMM is more persistent than the GMM throughout—but no single corrected ratio does.

The decoding channel separates cleanly. Holding standardization fixed at full-sample and switching the decoder from Viterbi to filtered cuts the HMM’s mean duration from 55.9 to 41.2 weeks—a 26% reduction attributable to decoding alone (channel L2), which brings the duration ratio to $4.6\times [2.6, 7.6]$. The smoothed decoder no longer sits between the two: under full-sample and expanding standardization it coincides with the Viterbi path to one decimal (identical spell counts), and only under rolling does it fall in between (17.2 weeks). This is what a highly persistent fit implies—when self-transition probabilities are large, the pointwise smoothed argmax rarely departs from the most likely path—and it means the practically relevant decode contrast is Viterbi versus filtered.

The mechanism is worth stating precisely, because it is easy to misread (Figure 2). Causal decoding does not repaint the timeline: only 51 weeks (3.3% of the sample) receive a different label under filtered versus Viterbi decoding, and the two decoders agree on 97.4% of Stressful assignments (filtered precision 94.4%, recall 98.1% against the Viterbi Stressful set). The decoders agree on which regime the market is in for almost every week. What they disagree on is when regimes switch, and because duration is a run-length statistic, a handful of relabelled weeks concentrated at regime boundaries fragments long runs into short ones, cutting the mean by a quarter. The look-ahead inflates the persistence statistic without much changing the regime assignment.

Transition entropy tells the same story through a second metric (Table 5). The GMM’s regime process is close to unpredictable, averaging 0.595 bits; the HMM under Viterbi is far more orderly at 0.136 bits, and causal filtering raises this only to 0.177 bits. The gap between the causal and non-causal decode is thus visible but modest—the fitted model’s orderliness is mostly genuine, not an artifact of hindsight—while the contrast with the GMM remains large under any decoder.

Table 5: Transition entropy per named regime (bits), full-sample standardization.

Labels	Calm	Transitional	Stressful	Mean
GMM	0.524	0.618	0.643	0.595
HMM Viterbi	0.100	0.130	0.180	0.136
HMM filtered	0.155	0.141	0.233	0.177

Table 6: Mean HMM regime duration (weeks): model-implied versus decoded. The analytic column is decode-invariant; the gap between it and the filtered decode measures channel L2, while variation down the analytic column reflects what each standardization does to the fit itself.

Standardization	Analytic $1/(1 - A_{ii})$	Viterbi	Filtered
Full-sample	58.2	55.9	41.2
Rolling-52	17.7	18.9	13.5
Expanding	60.3	50.4	37.8

The model-implied expected durations $1/(1 - A_{ii})$ locate the remaining variation in the fit itself (Table 6). Under full-sample standardization the HMM implies a mean duration of 58.2 weeks, which the Viterbi decode closely tracks (55.9 weeks);⁷ the filtered decode realizes about 70% of it (41.2). Since the analytic figure is decode-invariant, the gap between it and the filtered run-length is a clean measure of channel L2. The analytic column also resolves the rolling–expanding split: expanding standardization leaves the implied persistence essentially unchanged (60.3 weeks against 58.2 under full-sample), so removing future moments per se costs the model almost nothing, while rolling standardization cuts the implied persistence to 17.7 weeks—a reduction that comes from re-normalization redefining the features (Section 2.2), not from causality alone. Channel L1, in other words, is small when the causal scheme preserves the economic object and large when it changes it; channel L2 operates on top of whichever fit results.

4.3 Cross-model agreement

Despite operating on identical data and features, the two models substantially disagree. The adjusted Rand index between the GMM and HMM label sequences is 0.348 under Viterbi and 0.332 under filtered decoding—above chance, but far from strong concordance—and raw agreement on the named labels is 35.7%. The index is essentially unchanged across decoders, so the modest-agreement finding does not depend on the decoding choice.

It does depend on the period. Split at 2005, the index is 0.551 in the first half of the sample and 0.260 in the second: what agreement exists is concentrated in the earlier period. The two halves differ in episode composition—the first contains the dot-com cycle, the second the financial crisis and the COVID shock—and in how each model draws its calm–transitional boundary across the long post-2009 expansion, so I report the split as a robustness fact rather than press a structural explanation.

Agreement is not uniform across regimes. As the external validation below makes clearer, the models concur far more often during stress than during calm or transitional periods: the disagreement is concentrated where regime boundaries are inherently ambiguous. This is consistent

⁷The two are close but conceptually distinct: the analytic value is the geometric dwell time implied by the transition matrix, whereas the Viterbi value is the mean spell length of one particular decoded path.

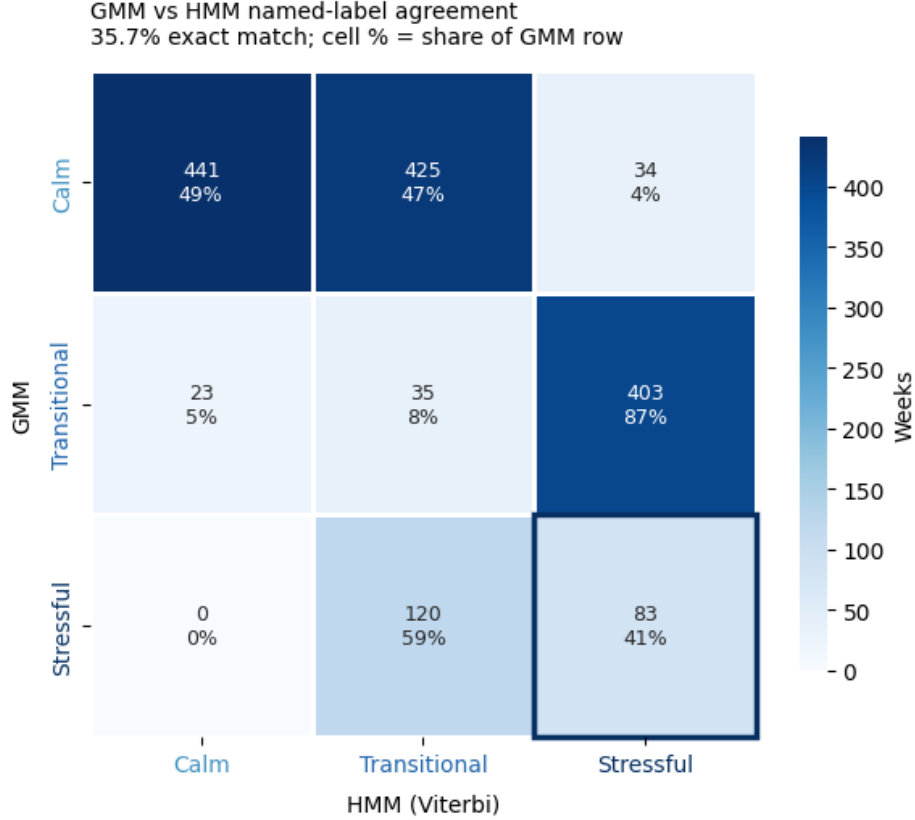


Figure 3: GMM versus HMM named-label agreement (35.7% exact match; cell percentages are shares of the GMM row).

with the two modelling assumptions—a static clusterer and a temporal model partition gradual transitions differently, while both readily identify acute stress.

Figure 3 shows the joint state frequencies. The GMM’s Calm weeks split nearly evenly between the HMM’s Calm (441 weeks) and Transitional (425); the GMM’s Transitional maps overwhelmingly to the HMM’s Stressful (403 of 461 weeks, 87%); and the GMM’s Stressful lands mostly in the HMM’s Transitional (120 of 203). The off-diagonal mass is therefore not random—it consists of one-regime offsets at adjacent boundaries, in both directions—while the Calm–Stressful corners are near-empty (34 weeks and none): the models never confuse calm with acute stress. Their disagreement is local.

4.4 External validity

The look-ahead corrections shrink the HMM’s apparent persistence; the question is whether they also erode the regimes’ economic content. Table 7 validates the Stressful regime against two external references—NBER recession dating and the $VIX > 20$ threshold—across the model and causal variants.

The pattern is no longer uniform, and saying so matters. For the full-sample and rolling configurations the Stressful label concentrates in recession weeks—52–61% of recession weeks against 9–32% of expansion weeks—and the concentration survives causal decoding intact: the filtered full-sample variant separates slightly better than its Viterbi counterpart. The expanding configu-

Table 7: External validation of the Stressful regime. Recession columns give the share of recession versus non-recession weeks labelled Stressful; the last column is Cohen’s κ against the $VIX > 20$ indicator.

Labels	Recession % Stressful	Non-recession %	$VIX > 20$ κ
GMM (pointwise)	52	9	0.35
HMM Viterbi (full)	55	31	0.55
HMM filtered (full)	58	32	0.56
HMM rolling + filtered	61	29	0.34
HMM expanding + filtered	44	35	0.49

ration is the weak case: 44% against 35%, with only 20% of dot-com recession weeks flagged. The same fit whose implied persistence stayed high in Table 6 spreads its Stressful label broadly across expansions, so under this scheme the label functions less as a recession indicator than elsewhere. Cohen’s κ against the volatility threshold spans 0.34–0.56 across configurations.

The rolling variant’s low κ (0.34) despite its strong recession concentration has a mechanical reading. Rolling standardization defines stress relative to the trailing year, while $VIX > 20$ is an absolute cut-off: a VIX of 22 after a placid year is locally extreme, the same level after a turbulent year is not, so decoupling from an absolute threshold is expected rather than anomalous. Consistent with this, the rolling variant flags 100% of COVID recession weeks—the relative measure saturates in the shock—while its concordance with the absolute rule stays low.

The two models recover stress differently. Against the VIX threshold, the GMM achieves 95% precision but only 32% recall: its pointwise Stressful label fires on the most extreme volatility spikes and misses sustained stress. The HMM trades precision for coverage (77% precision, 66% recall), absorbing broader stress blocks. The same contrast appears per episode: under the reference labels the GMM flags 30% of the dot-com downturn, 66% of the financial crisis, and 33% of the COVID shock as Stressful, against the HMM’s 52.5%, 53.7%, and 75%. The GMM sees crises as clusters of sharp spikes; the HMM sees them as regimes.

4.5 Regime economics

Table 8 reports the raw feature profile, sample share, and two drawdown statistics for each regime under the reference configuration. *Max DD (spell)* is the worst peak-to-trough decline of the S&P 500 within any single contiguous spell of the regime; *Mkt DD (in regime)* is the deepest marketwide underwater level—distance from the running peak—observed among weeks carrying the label. The distinction matters: only the first corresponds to a decline an investor could have experienced inside one visit to the regime.

Both models recover the expected monotone structure in volatility and returns—Calm regimes carry low VIX and positive drift, Stressful regimes elevated VIX and flat-to-negative returns—and within-spell drawdowns are monotone in stress for both, with the worst spell in each model’s Stressful regime reaching -46% , the core 2008–09 decline.

The cost of pointwise labelling is real, but it shows up elsewhere. The *Mkt DD (in regime)* column records how far below its running peak the market stood during weeks carrying each label: the GMM assigns Calm to weeks in which the market is as much as 41% underwater, against 25% for the HMM. A memoryless labeller reads low-volatility, positive-drift weeks near the bottom of a bear market as calm because, featurewise, they are; a temporal model carries the context. Together with the GMM’s 32% stress recall (Section 4.4), this is what having no temporal structure costs.

Table 8: Regime profiles under the reference configuration (full-sample standardization; HMM Viterbi). VIX and yield spread are raw means; return is the mean weekly return; share is the fraction of weeks; *Max DD (spell)* is the worst drawdown within any single contiguous spell of the regime; *Mkt DD (in regime)* is the deepest marketwide underwater level among weeks carrying the label.

Regime	VIX	Yield spread	Return	Share	Max DD (spell)	Mkt DD (in regime)
<i>GMM</i>						
Calm	14.8	1.13	+0.35%	58%	−7%	−41%
Transitional	23.8	0.29	−0.13%	29%	−26%	−29%
Stressful	31.8	2.09	−0.14%	13%	−46%	−56%
<i>HMM</i>						
Calm	13.3	0.48	+0.33%	30%	−7%	−25%
Transitional	18.8	1.93	+0.18%	37%	−16%	−46%
Stressful	26.3	0.44	−0.06%	33%	−46%	−56%

The middle states differ more than their shared name suggests. The GMM’s Transitional is a moderately volatile correction state (VIX 23.8, negative drift). The HMM’s Transitional is its largest regime (37% of weeks) and looks nothing like it: near-calm volatility (VIX 18.8), positive drift, the steepest yield curve in the table (1.93), and a market still as much as 46% below its prior peak—a post-stress *recovery* state, defined more by the yield curve than by volatility. The ascending-VIX naming rule calls it Transitional; economically it is the aftermath of stress. The GMM’s Stressful regime carrying the steepest spread on its side of the table (2.09) has the same reading: the curve steepens *during* recessions as policy rates fall, so inversion leads the cycle while steepening accompanies its trough.

4.6 Estimation is not unique

The channels of Section 3.3 concern information; this section concerns optimization. Baum–Welch maximizes a multimodal likelihood [10], so a single EM run returns one local optimum, and nothing guarantees it is a good one. To quantify the exposure, the full-sample HMM is refit from twenty seeds and each fit’s log-likelihood, decoded partition, and mean Viterbi duration are recorded (Table 11, Appendix A).

The spread is large. The best fit (log-likelihood −4,683.8; the baseline adopted throughout this paper, and also the best within the ten-seed restart set of Section 3.1) stands roughly 9 nats above the next optimum. Four seeds collapse to a degenerate fast-switching solution some 600 nats below, with mean regime durations of 2.3 weeks. Among the converged fits, decoded partitions differ from the selected baseline by adjusted Rand indices between 0.32 and 0.74, with mean Viterbi durations ranging from 41 to 58 weeks. Initialization, in other words, moves the decoded regimes by more than the decoding channel does. The single-seed fit used previously (seed 42, 86 nats below the optimum; Table 11) illustrates the stakes: under it, the decode correction of Section 4.2 measures 49% rather than 26%, and the expanding-window fit loses most of its implied persistence.

5 Discussion

5.1 The persistence gap is real, inflated, and configuration-dependent

The single result that most motivates choosing a temporal model over static clustering for regime detection is persistence: HMMs are reported to produce regimes that last on the order of economic cycles, where static methods flicker week to week. That contrast is real in direction and survives every configuration tested here, but no single number describes its size. Under the naive full-sample, Viterbi setup the HMM looks $6.2\times$ more persistent than the GMM. Causal decoding removes about a quarter of the HMM’s apparent duration—the share the Viterbi path reads off future observations. Causal standardization then either barely moves the fit (expanding, ratio $5.2\times$) or cuts it sharply (rolling, $2.2\times$), and Table 6 shows why: the rolling reduction comes from re-normalization redefining the features, not from causality per se. On top of all this, Section 4.6 shows that the fitted optimum itself shifts these figures by more than the decoding channel.

The reading is narrow and methodological, not a claim about utility: a regime duration is interpretable only once the standardization, the decoder, and the estimation protocol are specified. A 56-week figure obtained from full-sample standardization and Viterbi decoding describes what the regime structure looks like in retrospect; the duration an online system would have observed is about a quarter shorter, and under a rolling notion of stress it is shorter still. Reporting persistence without stating these choices—as informal comparisons typically do—conflates a retrospective description with an operational one.

5.2 Existence versus persistence

It would be a misreading to conclude that the look-ahead problem makes the regimes spurious. The direction of the external validation survives everywhere: in every configuration, recession weeks are more likely to be labelled Stressful than expansion weeks, and the causally decoded full-sample variant validates slightly better than its non-causal counterpart. But the strength of the validation is configuration-dependent in a way the earlier version of this paper did not acknowledge: the full-sample and rolling variants separate recessions sharply, while the expanding variant separates them weakly (44% against 35%) and misses most of the dot-com episode. The bias is still located mainly in the persistence statistic rather than in the regimes’ existence—but “existence” should be read as robust in direction, not invariant in strength.

Two caveats temper how much the regime structure can bear. First, the models substantially disagree: the adjusted Rand index sits near 0.33–0.35, positive but far short of the concordance one would want before treating a static and a temporal partition as interchangeable, and the agreement that exists is concentrated in market stress and in the first half of the sample. The choice of model materially changes the regimes outside of crises. Second, crisis identification is the robust, model-independent core of regime detection; the boundaries of calm and transitional periods are as much a product of modelling assumptions as a feature of the data. Claims of an objective regime partition should be read with that in mind.

5.3 An asymmetry, and a reusable lesson

The two models sit on opposite sides of a tradeoff the look-ahead analysis makes explicit. The GMM has no decoding step, so it is structurally immune to channel L2 and its durations were honest on that channel from the start—but this is a consequence of having no temporal model at all, and it carries a cost the corrected statistics now state precisely: the GMM fragments crises into short, sharp spikes, which produces its low stress recall (32%) and its willingness to label weeks

Calm while the market sits 41% below its peak (Sections 4.4 and 4.5). The HMM’s transition structure is exactly what lets it capture crises as coherent blocks, and exactly what exposes it to decoding look-ahead. The richer model buys genuine persistence and better stress coverage at the price of a channel that must be handled causally to be reported honestly.

This generalizes beyond the two models compared here. The three look-ahead channels—standardization, decoding, and parameter estimation—are present in any latent-state time-series model fit to financial data, they have different corrections, and they affect models asymmetrically, so collapsing them into a single notion of “look-ahead bias” obscures where the bias actually enters. To them this paper adds a companion of a different kind: initialization, whose failure mode is estimation variance rather than information leakage and whose remedy is restarts rather than causal protocols, but whose effect on the headline figures here exceeded that of the decoding channel. What is uncommon is not any ingredient but the execution—identical data and features, the channels separated, the estimation protocol surfaced, and the persistence claim corrected rather than asserted.

6 Limitations

The most consequential limitation concerns the one look-ahead channel left uncorrected. The analysis removes standardization look-ahead (L1) and decoding look-ahead (L2), but both models are still estimated once on the full sample (L3): every configuration in this paper—including the rolling-52, filtered variant, which is causal in its standardization and its decoding—rests on transition and emission parameters fit on the entire 1994–2023 period. Calling that variant “fully causal” is therefore shorthand for causal in the two channels this paper corrects; a genuinely end-to-end causal protocol would also re-estimate the models on a walk-forward or expanding basis and decode each week from parameters fit only on its past. That is the natural next step, and the corrections here should be read as making the *measurement* of persistence causal, not the estimation. Because L3 is common to both models, it does not bias the GMM–HMM contrast, but it does mean the absolute persistence and validity figures retain a component of in-sample fit.

Two further caveats attach to the numbers themselves. The restart protocol selects the best of ten seeded fits, and the diagnostic of Section 4.6 covers twenty; nothing guarantees a global optimum, although the selected fit is also the best across the wider diagnostic and the nearest competing optimum lies roughly 9 nats below it. And the bootstrap intervals for duration ratios treat the two models’ spell samples as independent, although both models are fit to the same weeks; the intervals should be read as approximate.

Relatedly, the rolling-52 variant should not be read as a pure bias correction of the naive configuration. Rolling re-normalization redefines each feature relative to its trailing year (Section 2.2), so its 2.2× ratio answers a related but different question—the persistence of *locally abnormal* conditions—from the 5.2× of the expanding variant, which preserves the economic object and removes only the future moments.

Relatedly again, the study is descriptive rather than predictive by design: it characterizes how modelling choices partition history and does not evaluate forecasting or out-of-sample generalization. The expanding-window scheme used here is a causal standardization, not a train/test split, and no claim is made that regimes estimated on early data predict later regimes.

Several modelling choices are fixed for comparability and bound the generality of the results. The number of regimes is set to three throughout; a different choice could split the Stressful regime into distinct crisis types or merge Calm and Transitional, and the three-regime structure should be read as imposed, not discovered. The feature set is deliberately compact—weekly equity

return, VIX, and the 10Y–2Y spread—so regimes are defined relative to those three dimensions; richer features (credit spreads, market breadth, realized higher moments) could shift the partition. Both models assume Gaussian emissions, a simplification for fat-tailed return data, and the HMM assumes a single stationary transition matrix over three decades despite plausible structural change in market dynamics across that span. Finally, the S&P 500 at weekly frequency proxies the broad market; the regimes need not transfer to small-cap, international, or fixed-income markets, or to intra-week dynamics.

7 Conclusion

This paper compared a static Gaussian mixture and a temporal hidden Markov model for unsupervised market-regime detection on identical weekly macro-financial data, in a pipeline that verifies every published number, in order to measure how much of the temporal model’s celebrated persistence survives causal measurement. The headline difference is real in direction but has no single corrected size. Causal decoding removes about a quarter of the HMM’s apparent regime duration, and the HMM-to-GMM duration ratio falls from $6.2\times$ under the naive configuration to $2.2\times$ under a rolling causal standardization but only to $5.2\times$ under an expanding one—the rolling scheme cuts deeper because it redefines the features, not merely because it is causal. A restart diagnostic adds that the fitted EM optimum moves these figures by more than the decoding channel: an earlier version of this study, resting on a single-seed fit 86 nats below the optimum, overstated the decode correction by a factor of two.

What survives is stated as precisely as what does not. The two models substantially disagree on the partition (adjusted Rand index near 0.33–0.35), with agreement concentrated in market stress and in the first half of the sample, so model choice materially shapes the regimes outside of crises. The regimes’ economic content—concentration of the Stressful label in NBER recessions and concordance with a $VIX > 20$ rule—holds in direction under every configuration, and in strength under the full-sample and rolling corrections, weakening under the expanding one. The look-ahead bias therefore lives mainly in the claimed persistence of the regimes, not in their existence. None of the components is novel; the contribution is a careful, reproducible execution in which the standardization, decoding, and estimation channels are separated, so that a familiar warning can be made precise: specify how a regime is standardized, how it is decoded—and how the model is estimated—before reporting how long it lasts.

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Disclosures

The author used Claude (Anthropic) for code review and verification, analysis, and manuscript editing, and used Cursor for code development with different models for different tasks, including GPT-5.5, Composer 2.5, and others. All analyses were independently reviewed by the author, who takes full responsibility for the content.

A Reproducibility

All results in this paper are pinned to an exact repository state and a fixed software environment, and the pipeline verifies itself—labels and numbers—before reporting anything.

A.1 Code and environment

The code, data, and archived regime labels are available at <https://github.com/nickdatak/Unsupervised-Discovery-of-Market-Regimes>, at commit 99cfc3b.⁸ The analysis runs under Python 3.9 with the pinned scientific stack in Table 9. Determinism rests on four fixed choices: the GMM uses `k-means++` initialization with ten restarts; the archived full-sample standardization is loaded rather than recomputed; the HMM is fit from the fixed seed tuple $\{42, \dots, 51\}$ with maximum-likelihood selection; and the spell bootstrap uses a fixed seed. Figures are produced with matplotlib and seaborn, which are left unpinned as they do not affect numerical results.

Table 9: Pinned environment (`requirements.txt`).

Component	Version
Python	3.9
numpy	1.26.4
pandas	2.1.4
scikit-learn	1.6.1
scipy	1.13.1
hmmlearn	0.3.3

⁸Full hash: 99cfc3b8b3e96c716e32851f6cad80ae55e76710

A.2 Data

All inputs are public: daily S&P 500 and VIX from Yahoo Finance and the 10Y–2Y spread from FRED. The merged weekly feature table (`market_features_weekly.csv`, 1,564 rows) and the three standardized variants—full-sample (1,564 rows; the archived baseline, not recomputed by the notebooks), rolling-52, and expanding (1,513 rows each)—are committed to `data/`, alongside the reference labels `gmm_regimes.csv` and `hmm_regimes.csv`. One caveat attaches to the claim that the dataset can be rebuilt from scratch: Yahoo Finance revises and backfills historical series, so a fresh download need not be bit-identical to the archived inputs. The committed CSVs, not the download recipe, carry the reproducibility claim.

A.3 Pipeline and gates

The analysis is a numbered notebook pipeline run in order, with all input and output routed through `regime_utils.py` (Table 10). Regimes are named per fit by ascending mean VIX (lowest → Calm, highest → Stressful), because the raw integer state IDs permute across standardization variants and seeds.

Verification operates at two levels. Before overwriting any archived labels, notebook 03 enforces a *label gate*: a fresh fit on the full-sample standardized features must match the archived CSVs at an adjusted Rand index of at least 0.99; both models clear it exactly, at $\text{ARI} = 1.000$. A *numbers gate* (`commit_table`) archives every table printed in this paper under `tables/` on first computation and requires every subsequent run to reproduce it to within 10^{-6} , so the published numbers are an exact, deterministic function of the committed data and the pinned environment.

Table 10: Notebook pipeline. Notebooks 01–05 are the analysis; 06 is a placeholder for the walk-forward refit discussed in Section 6 (not implemented); 07 and 08 generate figures.

Notebook	Function
01_features	Weekly (W-FRI) resampling, log returns, feature merge
02_standardize	Full-sample, rolling-52, and expanding standardizations
03_models	GMM and HMM fits; label gate; seed diagnostic
04_decode_durations	Decoders, durations, bootstrap, entropy, cross-model ARI
05_external_validation	NBER and VIX validation; regime profiles and drawdowns
06_robustness	Walk-forward refit (placeholder, not implemented)
07_decode_timeline	Filtered-versus-Viterbi regime timeline figure
08_vix_std_comparison	Standardization-comparison figure

A.4 Seed diagnostic

Table 11 reports the twenty-seed diagnostic of Section 4.6: for each seed, the total log-likelihood of the converged fit, the adjusted Rand index of its Viterbi partition against the selected baseline, and its mean Viterbi regime duration. Seeds 45, 52, 53, and 57 converge to a degenerate fast-switching optimum; the selected fit (seed 48) is the maximum across both the ten-seed restart set and the full diagnostic.

Table 11: Twenty-seed diagnostic, full-sample standardization, sorted by log-likelihood. ARI is against the selected baseline; duration is the mean Viterbi run-length in weeks.

Seed	Log-likelihood	ARI vs. baseline	Mean duration
48	-4683.8	1.000	55.9
59	-4692.5	0.385	47.4
61	-4692.5	0.385	47.4
60	-4692.5	0.385	47.4
49	-4708.4	0.420	41.2
44	-4728.3	0.320	50.5
47	-4728.3	0.320	50.5
51	-4728.3	0.320	50.5
54	-4728.3	0.320	50.5
55	-4728.3	0.320	50.5
56	-4730.1	0.334	53.9
58	-4737.0	0.621	47.4
43	-4740.2	0.715	57.9
46	-4740.5	0.737	57.9
50	-4769.8	0.549	50.5
42	-4769.9	0.552	50.5
45	-5291.8	0.356	2.3
57	-5291.9	0.355	2.3
52	-5292.1	0.354	2.3
53	-5292.1	0.354	2.3